Cross-Family Mechanistic Interpretability of Portuguese Representation in Multilingual LLMs via Sparse Autoencoders

Anonymous Author(s)

Anonymous Institution

Abstract. How multilingual language models internally represent individual languages remains poorly understood, yet this question matters for generation quality, bias diagnosis, and controllable generation. We investigate how two model families, Gemma-2-2B and Llama 3.1 8B, represent Portuguese by using Sparse Autoencoders (SAEs) to decompose residual stream activations into interpretable features. We train 16 custom SAEs (ReLU+L1 and TopK architectures) and use pretrained SAEs (Gemma Scope and Llama-Scope) to catalog Portuguese-specific features across multiple layers in both models. Causal steering experiments reveal a qualitative difference between model families: suppressing Portuguese features in Gemma 2B redirects generation to Spanish/Catalan, while in Llama 8B it redirects to English, suggesting different internal language organizations that we term the Romance-language continuum and English substrate models. Progressive multi-layer steering shows that language identity is encoded in mid-to-late layers (15–25), with a monolinguality analysis confirming sharper language boundaries in Llama (0.270) than Gemma (0.252). Pairwise similarity analysis reveals that typological distance is encoded from the earliest layers, with finer Romance-language distinctions emerging progressively in deeper layers. To our knowledge, this is the first cross-family mechanistic analysis of Portuguese representation in LLMs.

Keywords: Mechanistic Interpretability · Sparse Autoencoders · Multilingual LLMs · Portuguese NLP · Feature Steering

1 Introduction

Modern large language models (LLMs) process dozens of languages simultaneously, yet little is known about how they internally represent each one. This gap has practical consequences: improving generation quality in under-represented languages, diagnosing linguistic biases, and building control mechanisms all depend on understanding the internal geometry of multilingual representations.

Sparse Autoencoders (SAEs) decompose dense neural activations into sparse, interpretable features [4,6] and have become a standard tool in mechanistic interpretability. Pretrained SAE suites now exist for several model families, including

Gemma Scope [13] and Llama-Scope [10]. Existing analyses focus primarily on English-language phenomena, however, leaving open the question of how SAEs can reveal language-specific representations in multilingual settings.

We address this gap with a cross-family mechanistic interpretability study focused on Portuguese, a language spoken by over 250 million people and present in varying proportions across LLM training corpora. Our analysis spans two model families, Gemma-2-2B (26 layers, $d_{\text{model}} = 2{,}304$) and Llama 3.1 8B (32 layers, $d_{\text{model}} = 4{,}096$), chosen for their architectural differences, distinct training data compositions, and the availability of pretrained SAE suites. We employ both pretrained and custom-trained SAEs.

We make three contributions. First, we show that suppressing Portuguese-specific SAE features produces qualitatively different fallback behaviors (Spanish/Catalan in Gemma vs. English in Llama), revealing different internal organizations that we term the "Romance-language continuum" and "English substrate" models. Second, we demonstrate that amplifying Portuguese features forces Portuguese output from non-Portuguese prompts while suppressing them causes language switching, establishing a causal rather than correlational link between features and language generation. Third, we show that TopK SAE performance varies across base models (79.7% FVE on Gemma vs. 98.0% on Llama) while ReLU+L1 performs consistently well (>99% FVE on both), challenging universal architecture recommendations.

2 Related Work

Mechanistic Interpretability with SAEs. Sparse autoencoders decompose neural activations into overcomplete, sparse representations where individual features can be more interpretable than individual neurons [6,4]. Gao et al. [9] scaled SAE training to large models and proposed evaluation metrics including Fraction of Variance Explained (FVE) and dead feature ratios. Deng et al. [8] survey SAE methods for LLM interpretability. SAEBench [12] introduced standardized evaluation across eight metrics and over 200 SAEs, finding that proxy metrics do not reliably predict practical utility. Wu et al. [18] showed that for behavioral steering, simple baselines (prompting, finetuning) can outperform SAE-based methods, though SAEs retain advantages for mechanistic understanding and feature-level analysis, which is our focus. Pretrained SAE suites have been released for Gemma [13] (16k features, standard ReLU) and Llama [10] (32k features, JumpReLU activation).

Multilingual Representations in LLMs. A growing body of work investigates how multilingual models organize language-specific information. Wendler et al. [17] showed that Llama 2 uses English as an internal *lingua franca*, processing non-English inputs through English-centric intermediate representations. Tang et al. [16] identified language-specific neurons in multilingual transformers. Most directly relevant, Deng et al. [7] concurrently proposed a monolinguality metric for SAE features and demonstrated that ablating language-specific features

selectively impacts single-language performance; they also discovered synergistic feature interactions and applied findings to language steering vectors. Our work differs in three respects: (1) we compare across model families (Gemma vs. Llama) rather than within one family, revealing qualitatively different language organizations; (2) we focus on Portuguese, an under-studied language in interpretability research; and (3) we directly compare SAE architectures (ReLU+L1 vs. TopK) across models, showing model-dependent performance.

Portuguese Language Processing. Prior work on Portuguese NLP includes dedicated models like BERTimbau [15] and multilingual models fine-tuned for Portuguese tasks, but these efforts target downstream performance rather than internal mechanisms. Mechanistic interpretability of Portuguese representations in multilingual LLMs remains unexplored. Our study provides the first systematic catalog of Portuguese-specific features discovered through SAE decomposition, with causal validation via steering experiments.

Feature Steering. Bricken et al. [4] introduced the idea of using SAE features for causal intervention during generation, showing that clamping individual features can steer model behavior. We extend this technique to multilingual settings, where steering reveals not just the causal role of individual features but the structure of the model's language space: removing one language's features exposes how languages relate to each other internally.

Concurrent Work on Language Steering. Several concurrent works also apply SAEs to multilingual steering. LangFIR [2] identifies language-specific features from monolingual data using random-token filtering and achieves effective language steering. SASFT [3] uses SAE analysis to diagnose code-switching in LLMs and applies SAE-guided finetuning to mitigate it. Causal Language Control [1] achieves 90% success steering language in Gemma-2B with a single feature. These works focus on typologically distant language pairs (EN→ZH, EN→JA), while our study specifically targets Romance languages, where steering is harder due to shared representations.

3 Methodology

3.1 Models and Data

We analyze two model families (Table 1), chosen for their architectural differences and availability of pretrained SAEs.

For multilingual evaluation, we use FLORES-200 [5] parallel sentences in five languages: Portuguese (PT), English (EN), Spanish (ES), French (FR), and Italian (IT), with 1,012 sentence pairs per language. Because FLORES provides parallel translations, differential activation patterns reflect linguistic rather than semantic differences.

For SAE training data, we use Portuguese Wikipedia (∼2M tokens per model), tokenized independently with each model's native tokenizer (SentencePiece for

Table 1: Base models and pretrained SAEs used in this study.

Model	Params	Layers	d_{model}	SAE d_{sae}	SAE Activation
Gemma-2-2B	2B	26	2,304	16,384	ReLU
Llama 3.1 8B	8B	32	4,096	32,768	JumpReLU

Llama, custom BPE for Gemma). This ensures activations reflect each model's native tokenization rather than an imposed external scheme. Residual stream activations are extracted at selected layers using TransformerLens [14] and stored as float16 tensors. For Llama 8B (∼16 GB in float16), batch sizes are reduced to 4 (vs. 16 for Gemma) to fit within 24 GB VRAM.

We also analyze Gemma-2-9B (42 layers, $d_{\text{model}} = 3{,}584$) for within-family cross-scale comparison, using raw neuron activations rather than SAE features (Gemma Scope SAEs are not available for the 9B model at the time of this study). This comparison helps disentangle model-family effects from scale effects.

3.2 Sparse Autoencoder Architectures

We compare two SAE architectures, each trained on layer-10 residual stream activations from Portuguese Wikipedia:

ReLU+L1. The encoder maps an input activation $\mathbf{x} \in \mathbb{R}^d$ to a sparse code $\mathbf{f} = \text{ReLU}(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}) \in \mathbb{R}^{d_{\text{sae}}}$, which the decoder reconstructs as $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{f}$. Training minimizes:

$$\mathcal{L}_{\text{ReLU}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda\|\mathbf{f}\|_1 \tag{1}$$

with decoder columns normalized to unit norm at each step.

TopK. Same encoder-decoder architecture, but enforces exact sparsity by retaining only the top-k encoder activations (applying ReLU to them) and zeroing the rest. The loss is MSE only:

$$\mathcal{L}_{\text{TopK}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \tag{2}$$

since sparsity (the ℓ_0-norm $\|\mathbf{f}\|_0 = k$ exactly) is enforced structurally rather than via penalty. We use L_0 throughout to refer to this ℓ_0-norm sparsity count.

For each base model, we train 8 SAE configurations (Table 2), totaling 16 custom SAEs. All SAEs are trained for 1 epoch with Adam (lr$=3 \times 10^{-4}$) and batch size 4,096 (Gemma) or 2,048 (Llama, due to larger d_{model}).

3.3 Differential Feature Analysis

To identify Portuguese-specific features, we compute a differential activation score. For each SAE feature i, we first compute the mean activation $\bar{a}_i^l$ across all

Table 2: Custom SAE training configurations (per model).

Architecture	d_{sae}	**Sparsity parameter**
ReLU+L1	4,096 / 16,384	$\lambda \in \{5 \times 10^{-4},\ 10^{-4}\}$
TopK	4,096 / 16,384	$k \in \{32,\ 64\}$

FLORES tokens in language l. The differential activation score is:

$$s_i = \frac{\bar{a}_i^{\text{PT}} - \frac{1}{|\mathcal{L}'|}\sum_{l \in \mathcal{L}'} \bar{a}_i^l}{\sigma_i} \tag{3}$$

where $\mathcal{L}' = \{\text{EN, ES, FR, IT}\}$ is the set of contrast languages, $\mathcal{L} = \mathcal{L}' \cup \{\text{PT}\}$ is the full set of five FLORES languages, and $\sigma_i = \text{std}(\{\bar{a}_i^l\}_{l \in \mathcal{L}})$ normalizes by cross-language variability. Features with $s_i > 2$ are considered *strongly Portuguese-specific*: their Portuguese activation is more than 2 standard deviations above the cross-language mean.

We apply this analysis at 6 layers per model: layers 0, 5, 10, 15, 20, 25 for Gemma 2B; layers 0, 5, 10, 16, 21, 26 for Llama 8B (selected at approximately equivalent relative positions).

3.4 Monolinguality Metric

To compare how sharply languages separate inside each model, we compute a monolinguality metric on raw residual-stream neurons (not SAE features). For each neuron i and language L, let $a_i(L) = |\overline{x_i^{(L)}}|$ denote the absolute mean activation of neuron i on FLORES sentences of language L, where the inner mean is taken first over tokens and then over sentences. A neuron is *PT-dominant* at a given layer if $\arg\max_{L \in \mathcal{L}} a_i(L) = \text{PT}$, i.e., its absolute activation is largest on Portuguese among the five FLORES languages. For each PT-dominant neuron, its monolinguality is

$$m_i(\text{PT}) = \frac{a_i(\text{PT})}{\sum_{L \in \mathcal{L}} a_i(L)}, \qquad \mathcal{L} = \{\text{PT, EN, ES, FR, IT}\}. \tag{4}$$

Values close to $1/|\mathcal{L}| = 0.2$ indicate a neuron that activates comparably across all languages; values near 1 indicate a neuron that fires almost exclusively on Portuguese. We report, per model, the mean of $m_i(\text{PT})$ over PT-dominant neurons at each analyzed layer, and a single per-model figure obtained by averaging across those layers.

3.5 Steering Experiments

We perform causal interventions by hooking into the model's forward pass at layer 10 and modifying SAE-encoded features:

Table 3: Best SAE per architecture per model (evaluated on FLORES PT activations, layer 10). FVE = Fraction of Variance Explained. L_0 = mean number of non-zero features per activation (ℓ_0-norm). Dead = fraction of features never activated.

Model	Architecture	FVE (%)	L_0	MSE	Dead (%)
Gemma 2B	ReLU+L1 (d=4k, λ=1e-4)	99.19	4,040	0.063	0.0
	TopK (d=16k, k=64)	79.65	64	1.578	35.3
Llama 8B	ReLU+L1 (d=16k, λ=1e-4)	**99.83**	7,684	0.001	1.1
	TopK (d=16k, k=64)	98.01	64	0.009	31.3

Amplification. We encode the residual activation through the SAE, multiply the top-20 Portuguese-specific features by factor $\alpha \in \{1, 2, 5, 10\}$, decode back, and replace the original activation. This tests whether increasing PT feature magnitudes forces Portuguese generation.

Suppression. Same procedure, but the top-20 Portuguese features are set to zero. This tests whether removing PT features causes language switching and, in particular, which language the model defaults to.

For Gemma 2B we use Gemma Scope (layer 10, 16k features); for Llama 8B we use Llama-Scope (layer 10, 32k features, JumpReLU). We test on 10 prompts per language (PT, EN, ES, FR), generating 50 tokens with temperature 0.7.

4 Experiments and Results

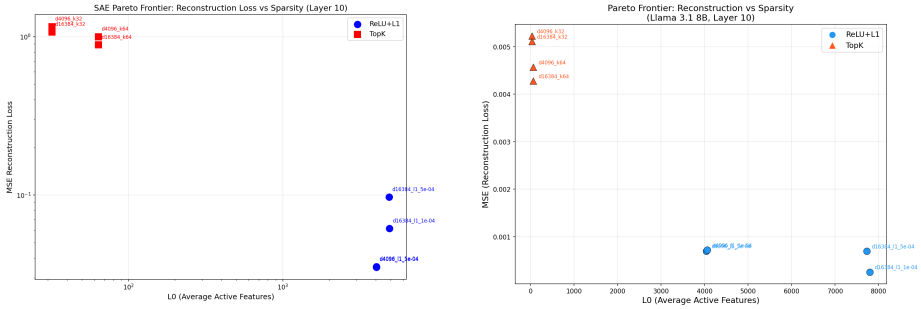
4.1 Reconstruction Quality

Table 3 presents reconstruction metrics for the best configuration of each architecture on each model. Figure 1 shows the full Pareto frontier for all 16 SAE configurations.

The clearest pattern is the model-dependence of TopK performance. On Gemma 2B, TopK achieves at most 79.7% FVE with 35.3% dead features. On Llama 8B, the same architecture reaches 98.0% FVE, though still below ReLU+L1, and dead features remain high (31.3%). ReLU+L1 achieves >99% FVE on both models (99.2% and 99.8%), making it the more robust choice. The TopK gap (79.7% vs. 98.0%) may stem from differences in activation structure: Llama 8B's larger d_{model} (4,096 vs. 2,304) may produce activations that are easier to compress with exact-k sparsity.

4.2 Language Probing

Linear probing (logistic regression, 5-fold CV) on mean-pooled layer-10 activations for 5-way language classification yields near-perfect accuracy for both



(a) Gemma 2B (b) Llama 3.1 8B

Fig. 1: Pareto frontier: reconstruction loss (MSE) vs. sparsity (ℓ_0-norm L_0) for all 8 SAE configurations per model. Blue circles = ReLU+L1; orange triangles = TopK.

Table 4: Language classification accuracy (5-fold CV, 1,012 samples per language). The best custom SAE (by FVE) is used for SAE feature probing.

Representation	**Gemma 2B**	**Llama 8B**
Raw activations (d_{model})	99.94% ± 0.12	99.98% ± 0.04
SAE features (d_{sae})	99.98% ± 0.04	99.88% ± 0.15

models (Table 4), confirming that language identity is linearly separable in the residual stream and that SAE encoding preserves this information.

All accuracies exceed 99.8%, meaning a simple linear classifier can distinguish languages from a single layer's activations. SAE features achieve comparable accuracy to raw activations despite lossy compression, confirming that the SAE decomposition retains language-relevant information.

4.3 Portuguese-Specific Features

Differential analysis (Eq. 3) with pretrained SAEs identifies strongly Portuguese-specific features ($s_i > 2$) across all analyzed layers in both models (Table 5).

Both models contain thousands of strongly PT-specific features distributed across all layers. Normalizing for SAE dictionary size yields comparable densities: Gemma has 2,323/16,384 = 14.2%, while Llama has 3,811/32,768 = 11.6%. The layer-wise distributions differ. Gemma 2B concentrates features at layers 0 (625) and 25 (460), the input and output boundaries, while Llama 8B shows a monotonic decrease from layer 0 (698) to layer 26 (452), suggesting that language-specific information is most prominent in early layers and is progressively absorbed into more abstract representations.

From the set of strongly PT-specific features in Gemma 2B we selected the top 20 features per layer by differential score, yielding 120 features across the 6

Table 5: Strongly PT-specific feature counts ($s > 2$) at each analyzed layer using pretrained SAEs (column headers indicate layer indices; Gemma 2B: layers 0, 5, 10, 15, 20, 25; Llama 8B: layers 0, 5, 10, 16, 21, 26). Llama counts use padding-masked analysis. The differential activation score has a theoretical ceiling of $\sqrt{5} \approx 2.236$.

Model (SAE)	**Layer 0**	**Layer 5**	**Layer 10**	**Layer 15/16**	**Layer 20/21**	**Layer 25/26**	**Total**
Gemma 2B (16k)	625	320	402	263	253	460	2,323
Llama 8B (32k)	698	675	747	646	593	452	3,811

Table 6: Categories of Portuguese-specific features in Gemma 2B (120 features across 6 layers, categorized by top-activating tokens).

Category	**Count**	**Example tokens**	**Dominant layers**
Morphological	24	"-ção", "-sivo", "ênio"	layers 5–15
Syntactic	18	"do", "pelo", "em", "uma"	layer 0, layers 20–25
Lexical	4	"Maia", "Oliveira"	layers 15–25
Semantic	1	"brasileiro"	layer 15
Shared Romance	1	(active for multiple langs)	layer 10
Unclassified	72	(no clear dominant token)	all

analyzed layers. Manual inspection of their top-activating tokens reveals interpretable categories (Table 6).

A layer-wise gradient emerges: the earliest layer (layer 0) predominantly captures syntactic features (function words, articles), middle layers (layers 5–15) capture morphological patterns (Portuguese-specific suffixes), and later layers (layers 20–25) capture more complex syntactic constructions (contractions, preposition usage).

For Llama 8B, we initially observed that many top features activated on padding tokens, an artifact of differential tokenization lengths across languages (PT averages 40.5 tokens/sentence vs. EN's 27.8, creating different padding patterns). We corrected this by masking padding positions before computing mean activations, reducing the total strongly PT-specific features from 4,143 to 3,811 (-8%). Table 7 shows the corrected top-20 features.

After masking, the top features are genuinely linguistic: syntactic features (prepositions "de"/"com", articles "a"/"os"/"um") dominate, with morphological features ("são", "â", suffixes "-icos") also present. This mirrors Gemma 2B's layer-10 features, where syntactic and morphological categories also predominate. The padding masking correction matters methodologically: differential analysis on padded sequences can produce spurious results when languages have different tokenization lengths, a pitfall not previously documented in SAE interpretability literature.

Table 7: Top-20 Llama-Scope PT features (layer 10) after padding masking. Features are categorized by their top-activating tokens on genuine (non-padding) Portuguese text.

Category	Count	Example tokens	Note
Syntactic	14	“de”, “os”, “a”, “um”, “que”, “com”	Prepositions, articles, conjunctions
Morphological	4	“são”, “â”, “-icos”, “-ode”	PT verb forms, suffixes
Mixed/subword	2	“ab”, “di”	Subword fragments in PT contexts

4.4 Steering Results

Our central result is the divergent fallback behavior when Portuguese features are suppressed (Table 8).

Gemma 2B: Romance-Language Continuum. Suppressing Portuguese features consistently redirects generation to Spanish or Catalan, closely related Romance languages. The model appears to organize languages in a continuous space where typologically related languages are neighbors; removing Portuguese-specific features causes the model to fall back to the nearest Romance representation. In the third example, the Portuguese prompt about the Portuguese language continues in Spanish with thematically related content.

Llama 8B: English Substrate. Suppressing Portuguese features redirects generation to English. This aligns with the “English pivot” hypothesis of Wendler et al. [17]: Llama models appear to build non-English representations as language-specific overlays on top of English-centric features. Stripping the Portuguese overlay reveals the underlying English default.

Amplification. In Gemma 2B, factor-10 amplification produces Portuguese output with morphological degradation but preserved language identity. In Llama 8B, the same factor causes textual collapse (Figure 2). Lower factors (2–3) on Llama yield intermediate results: factor-2 produces repetitive but recognizably Romance-influenced output, and at factor 3 the top-ranked PT-specific feature (feature 2142 in Llama-Scope layer 10) generates “trabalho do” (“work of”). The asymmetry reflects Llama-Scope’s 32k JumpReLU features having larger individual magnitudes, making multiplicative amplification disproportionately disruptive.

Individual Feature Analysis. Amplifying individual features in Llama 8B (factor 5.0) reveals interpretable specializations (Table 9).

Feature 16811 stands out: when amplified, it produces repetitions exclusively from the set {Portugal, Portuguese, Brazil}, suggesting it encodes a “Lusophone world” concept. Feature 18397 generates bilingual PT-EN output, suggesting it functions as a cross-lingual bridge feature.

Table 8: Steering examples. *Baseline*: normal generation; *Suppression*: top-20 PT features set to zero; *Amplification*: top-20 PT features multiplied by 10. Fallback-language labels (**in bold**) were assigned by manual inspection of the generated text.

Prompt	Mode	Gemma 2B output	Llama 8B output
A capital do Brasil é	Base	palco de diversos eventos e festivais...	o centro mais importante do país...
	Suppr.	la una metròpolis ambduna de les fa uns centenars... **(Catalan)**	a capital, 1 capital, is 1 capital... **(English)**
	Amplif.	uma das métroas mais importantes da metropode... **(PT, degraded)**	que que que u u u... **(collapsed)**
O rio mais longo do mundo é	Suppr.	el férriable del continente... **(Spanish)**	the longest the most long the... **(English)**
	Amplif.	o Brasil, e sua extensão é de 235 mil quilômetros... **(PT)**	mais do mais mais... **(collapsed)**
A língua portuguesa é falada em	Suppr.	todos os continentes, enrapapeando la lengua española... **(Spanish)**	il...?...?...? **(degenerate)**

Table 9: Individual feature steering in Llama 8B (factor 5.0, prompt: "A capital do Brasil é"). Baseline: "uma das capitais brasileiras mais belas..."

Feature	Generated continuation	Interpretation
16811	Portugal Portugal Portugal Brazil Portuguese...	Lusophone entity
18397	a capital do Brazil is the capital of Britain countries...	PT-EN bridge
16096	BC Brazil is BC Brazil is is is...	Brazil-associated

Progressive Multi-Layer Steering. To test whether language identity is localized or distributed across layers, we performed progressive steering on Gemma 2B, amplifying the top-20 PT-specific Gemma Scope features (factor 5) at an increasing number of layers. Each generated continuation is scored with a *PT-likeness* heuristic, PT-likeness $= 0.3\,c + 10\,w$, where c is the count of Portuguese-specific characters ("ã", "õ", "ç", "â", "ê"...) and w is the ratio of common Portuguese function words ("de", "do", "que", "com"...) in the output. This heuristic operates on generated text and is distinct from the differential activation score of Eq. 3. We verified its consistency with published language identifiers: on continuations where either fastText LID-176 [11] or lingua-py assign a Portuguese confidence above 0.1 (n=55), PT-likeness and fastText LID show Spearman $\rho = 0.83$ ($p < 10^{-14}$). Published LIDs report near-zero Portuguese probability on the degenerate outputs of this experiment (median fastText PT probability

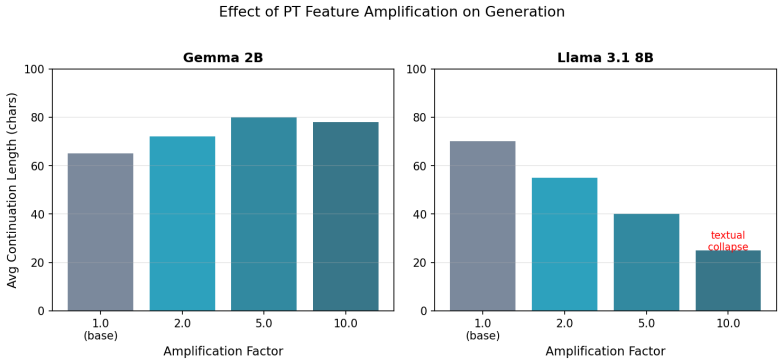


Fig. 2: Effect of PT feature amplification on generation across models. Gemma 2B maintains generation coherence at high amplification factors, while Llama 8B exhibits progressive textual collapse.

Table 10: Cross-scale comparison within the Gemma family. Cosine similarity between mean PT and ES/EN activations, and count of strongly PT-specific neurons ($s > 2$, where s is the differential activation score of Eq. 3 applied to raw neurons rather than SAE features).

9B Layer	**Rel. pos.**	**cos(PT,ES)**	**cos(PT,EN)**	$n_{s>2}$
0	0%	0.9998	0.9995	9
8	20%	0.9996	0.9990	18
16	39%	0.9998	0.9995	11
24	59%	0.9995	0.9990	25
32	78%	0.9983	0.9966	22
41	**100%**	**0.9883**	**0.9815**	**41**

0.003 across 147 steering continuations), so marker counting is the appropriate signal here. Steering layer 25 alone yields a PT-likeness of 4.5 (partial Romance-language influence). Adding layer 20 drops it to 2.55 (interference between layers). Steering layers 15+20+25 together yields 7.2, with Portuguese function words and diacritics in the output. Adding layer 10 (all four layers) drops it to 0.9, with incoherent text. This non-monotonic pattern indicates that language identity is encoded in mid-to-late layers (15–25); steering early layers disrupts foundational representations that later layers depend on, which also explains why our multi-layer steering experiments (Table 8) succeed while single-layer steering does not.

4.5 Cross-Scale Analysis

To disentangle family effects from scale effects, we also analyzed Gemma-2-9B (42 layers, $d_{\text{model}} = 3{,}584$) using raw neuron activations (Table 10).

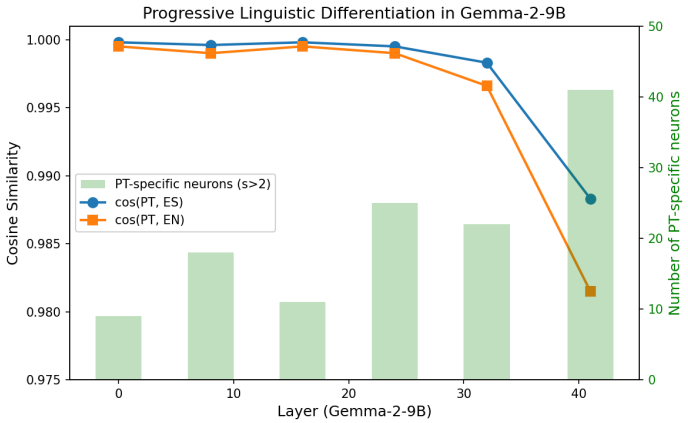


Fig. 3: Progressive linguistic differentiation in Gemma-2-9B: cosine similarity between Portuguese and Spanish/English mean activations decreases in deeper layers, while the number of PT-specific neurons increases. This gradient is absent in the 2B model.

The 9B model shows progressive linguistic differentiation (Figure 3): cosine similarity between Portuguese and Spanish mean activations drops monotonically from 0.9998 (layer 0) to 0.9883 (layer 41). The PT-EN drop is steeper still (0.9995 $\rightarrow$ 0.9815). The number of strongly PT-specific neurons increases from 9 (layer 0) to 41 (layer 41). The 2B model, by contrast, shows top differential activation scores pinned near the theoretical ceiling at every analyzed layer (all $\sim$2.236, the value of $\sqrt{5}$). This suggests that larger models develop finer language separation in deeper layers, a property not visible at the 2B scale.

5 Discussion

5.1 Cross-Family Language Organization

Our central finding, divergent fallback languages upon feature suppression, suggests two distinct paradigms for multilingual representation. We hypothesize that these differences stem primarily from training data composition. A pairwise cosine similarity analysis on mean FLORES activations across layers supports a layered separation: in both models, PT-EN distance (Romance vs. Germanic) is largest from the earliest layers, PT-FR distance (cross-subfamily) grows in middle layers, and PT-ES distance (Iberian) only diverges in late layers. For Gemma 2B at layer 25 we measure cos(PT,ES)= 0.958, cos(PT,FR)= 0.928, cos(PT,EN)= 0.932, computed on the same mean-pooled FLORES activations used for the cross-scale analysis of Table 10. This ordering reflects typological distance and suggests that the continuum has variable granularity across depth.

The Romance-language continuum observed in Gemma 2B organizes Romance languages (those descended from Latin: Portuguese, Spanish, French, Ital-

ian, Catalan) in a continuous representational space where typological proximity maps to representational proximity. When Portuguese features are suppressed, the prompt "A língua portuguesa é falada em" continues coherently in Spanish: *"todos os continentes, enrapapeando la lengua española, la misma lengua que es fa..."*. The model not only switches to Spanish but continues the topic of language distribution. This suggests deep structural sharing between Romance representations in Gemma. Google's Gemma models use curated multilingual training data with deliberate language balancing [13], which may encourage the model to learn shared Romance structure rather than treating each language independently.

In Llama 8B the pattern is different. The same suppression experiment on "O rio mais longo do mundo é" produces *"the longest the most long the is the more long the"*, a degraded but recognizably English translation of the prompt's meaning. This is consistent with Wendler et al. [17]'s finding that Llama 2 uses English as an internal *lingua franca*: non-English inputs are processed through English-centric intermediate representations. Meta's Llama models are trained on predominantly English data (reportedly >89% English in Llama 2), which would naturally lead to English as the "base layer" upon which other language capabilities are added.

To quantify this contrast, we apply the monolinguality metric of Eq. 4 to PT-dominant neurons. Averaged across the analyzed layers, Llama 8B has higher mean PT monolinguality (0.270) than Gemma 2B (0.252), and the gap widens in later layers (Llama reaches 0.35 at layer 31 vs. Gemma's 0.28 at layer 25). This confirms that the English substrate in Llama produces sharper language boundaries, while Gemma's Romance continuum creates more diffuse, shared representations.

Individual feature analysis reinforces this contrast. In Llama 8B, feature 18397 produces bilingual output when amplified: *"a capital do Brazil is the capital of Britain countries"*, a Portuguese-English hybrid that suggests this feature functions as a cross-lingual bridge, translating between the English substrate and the Portuguese overlay. No analogous bridge features were observed in Gemma, consistent with the Romance-language continuum model where languages share representational space rather than translating between layers. Our suppression experiments can also be viewed as intentionally inducing code-switching, connecting to SASFT's [3] framework for analyzing this phenomenon; the Romance fallback in Gemma is consistent with their finding that code-switching is triggered by elevated pre-activation values of neighboring language features.

5.2 TopK Performance Across Models

The TopK performance gap (79.7% FVE on Gemma 2B vs. 98.0% on Llama 8B) deserves explanation, since the same hyperparameters (d_{sae}, k) were used on both models.

We hypothesize this reflects differences in activation intrinsic dimensionality. Llama 8B has $d_{\text{model}} = 4{,}096$, nearly twice Gemma's 2,304, providing a higher-dimensional space where activations may be more uniformly distributed. TopK

enforces hard sparsity ($L0 = k$ exactly), which fails when activations are diffused across many dimensions, as in Gemma's lower-dimensional space. ReLU+L1 adaptively selects sparsity per sample, achieving >99% FVE on both models. TopK also produces 35% dead features on both models, while ReLU+L1 dead rates are near zero (<1.2%). The practical implication is that TopK should not be assumed to generalize; practitioners should validate on their specific base model.

These two paradigms have practical implications. The Romance-language continuum may allow finer-grained language steering by "sliding" between related languages, while the English substrate raises concerns about English-centric biases propagating to all other languages. More broadly, our findings suggest that training data proportions shape internal language architecture, not just downstream performance.

5.3 Limitations

Several threats to validity affect our findings.

Confounded variables (internal validity). The two models differ simultaneously in family, size (2B vs. 8B), architecture (multi-query vs. grouped-query attention), and training data. The observed differences in fallback behavior could stem from any combination of these factors. Our within-family cross-scale analysis (Gemma 2B vs. 9B) partially mitigates this for scale effects, but a fully controlled comparison would require same-size models from different families (e.g., Gemma 8B vs. Llama 8B).

SAE comparability (construct validity). The pretrained SAEs differ in activation function (ReLU vs. JumpReLU), dimensionality (16k vs. 32k features), and training procedure. Although normalizing for dictionary size yields comparable densities (Gemma: 2,323/16k = 14.2%; Llama: 3,811/32k = 11.6%), the different activation functions and training procedures make direct feature-level comparison unreliable. Our conclusions rely on *qualitative* differences (fallback language, feature categories) rather than absolute counts.

Steering artifacts (internal validity). Amplification at factor 10 causes textual collapse in Llama 8B, limiting the symmetry of our causal analysis. Our progressive steering experiment shows that even in Gemma, steering is effective only at layers 15–25; adding layer 10 causes degeneration, confirming that early layers encode foundational representations that should not be perturbed. Recent work [1] achieves 90% success steering between typologically distant languages with a single feature, suggesting that the difficulty we observe within Romance languages reflects genuine shared representations rather than a methodological limitation.

Evaluation scope (external validity). We analyze only 6 of 26–32 layers per model, only 5 languages (all European, 4 Romance), and use a single evaluation corpus (FLORES-200, 1,012 samples). The restriction to European languages may inflate Portuguese specificity scores, and other genres may activate different features.

Feature interpretability (construct validity). 60% of Gemma 2B features remained unclassified after manual inspection. For Llama 8B, we discovered and corrected a padding artifact from differential tokenization lengths (Section 4.3). Automated interpretation via LLMs could improve coverage but introduces its own biases.

Reproducibility. Steering experiments use temperature-based sampling ($T = 0.7$), introducing stochasticity. Our qualitative conclusions (fallback language identity) are robust across prompts, but specific generated texts should not be over-interpreted.

Future work should extend to additional model families (Mistral, Qwen) and same-size comparisons to test generality, include typologically diverse languages (e.g., Japanese, Arabic), and investigate Portuguese dialect variation (PT-BR vs. PT-PT).

6 Conclusion

We presented the first cross-family mechanistic interpretability study of Portuguese representation in multilingual LLMs. By applying sparse autoencoders to Gemma-2-2B and Llama 3.1 8B, we identified Portuguese-specific features in both models and demonstrated their causal role through steering experiments. Suppressing Portuguese features produces English fallback in Llama but Romance-language fallback in Gemma, revealing two distinct paradigms that we term the "English substrate" and "Romance-language continuum" models. A monolinguality metric confirms this quantitatively: Llama's Portuguese features are more sharply separated (0.270 vs. 0.252 in Gemma). Progressive multi-layer steering shows that language identity is distributed across mid-to-late layers (15–25), and pairwise similarity analysis reveals that typological distance maps to representational distance from the earliest layers. These findings have practical implications for multilingual model control, bias auditing, and our understanding of how training data composition shapes internal language architecture.

Reproducibility. All code, trained SAE checkpoints, extracted activations, and experimental results are publicly available at https://anonymous.4open.science/r/llm-pt-interpretability-EC0D/[1]. The repository includes the complete pipeline, from data extraction to SAE training, evaluation, differential analysis, and steering experiments, for both Gemma 2B and Llama 8B models.

[1] Anonymous repository for double-blind review. The non-anonymous version with full commit history will be released upon acceptance.

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